

Vendor Quality and Logistic Scorecard

Problem Statement

On an e-commerce marketplace, customers judge the platform based on their overall shopping experience, not just the individual seller. Late deliveries and poor service from a small group of sellers can damage customer trust, reduce repeat purchases, and hurt the platform's reputation. This project builds a seller performance scorecard that identifies underperforming sellers using delivery, review, and sales data. It helps the operations team measure seller performance, prioritize improvement efforts, and make data-driven decisions.

Questions answered: Which sellers deliver late most often? Which earn the worst reviews (with enough volume to be meaningful)? Which categories drive revenue, lateness, and poor ratings? Is delivery performance improving over time? Who should be coached, monitored, or offboarded?

Dataset

Olist Brazilian E-commerce (Kaggle) — 5 tables, ~100K orders

Link: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce/data>

Headline Results

- 1.Orders are delivered about **12 days** earlier than the estimated delivery date on average. However, this average hides a small group of sellers with poor performance, where 30–64% of orders are delivered late.
- 2.The lowest-performing qualified seller scored **71.8/100**, with an average review rating of **1.93** and **64%** of orders delivered late across 14 orders. Another seller, ranked third from the bottom, handled **108 orders** with an average rating of 2.27, making them one of the biggest contributors to poor customer experience.
- 3.A seller can still deliver before the estimated date on average while being late on one out of every three orders. For this reason, the scorecard uses late delivery percentage instead of average delivery delay to measure performance.
- 4.Late deliveries vary by product category. Portable kitchen appliances, watches & gifts, and telephony have the highest late delivery rates (around 6–8%), while the best-performing categories are closer to 5%.
- 5.Health & beauty is the highest-revenue category, generating about **R\$1.2 million** in sales. Books and CDs/DVDs receive the highest customer ratings (above 4.4), and São Paulo accounts for the largest share of sales.

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Python pipeline

WHAT	WHY
Load 5 CSVs; audit shape, datatypes, missing values per table	Know what needs fixing before touching anything
Products: category nulls → 'Unknown'; numeric nulls → 0	Keeps rows usable without breaking numeric datatypes
Orders: keep only delivered ; drop null dates; parse datetimes	Delivery metrics are meaningless for canceled/processing orders
Engineer <code>delivery_time_days</code> and <code>sla_delay_days</code>	The two columns powering every dashboard metric (late %, risk score)
Reviews: Kept only the most recent review for each order by removing duplicate review submissions. This affected 547 orders where customers had submitted more than one review.	One customer must never count twice in a seller's rating
Filter <code>order_items/reviews</code> to surviving orders	Referential integrity — no orphan rows, no FK failures
Load to MySQL, parents before children	Foreign keys reject child rows whose parents don't exist yet

MySQL layer

Designed a normalized 5-table database schema with primary keys for dimension tables, a composite primary key for the order items fact table, and foreign keys to maintain referential integrity.

Created a flat analysis view for reporting and built the seller scorecard query with a minimum threshold of 10 orders, ensuring low-volume sellers don't skew the rankings.

Issue fixed during development:

The initial late delivery calculation had a grain mismatch. It counted late item rows in the numerator but distinct orders in the denominator. As a result, orders containing multiple items could be counted more than once, causing late delivery rates to exceed 100%.

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This was corrected by calculating late deliveries at the order level using `COUNT(DISTINCT CASE WHEN is_late THEN order_id END)`, ensuring both the numerator and denominator use the same grain. The same approach was applied to the 1-star review rate calculation.

Power BI measures

- `Total Orders = DISTINCTCOUNT(fact[order_id])`
- `Avg Review Score = AVERAGE(fact[review_score])`
- `Late Deliveries % = DIVIDE(CALCULATE(DISTINCTCOUNT(fact[order_id]), fact[is_late] = 1), [Total Orders])`
- `Vendor Risk Score = ROUND(0.6 * ((5 - [Avg Review Score]) / 4 * 100) + 0.4 * ([Pct Late Deliveries] * 100), 1)`
- `Is Qualified Seller = IF([Total Orders] >= 10, 1, 0)`
- `Risk Score (Qualified) = IF([Is Qualified Seller] = 1, [Vendor Risk Score])`

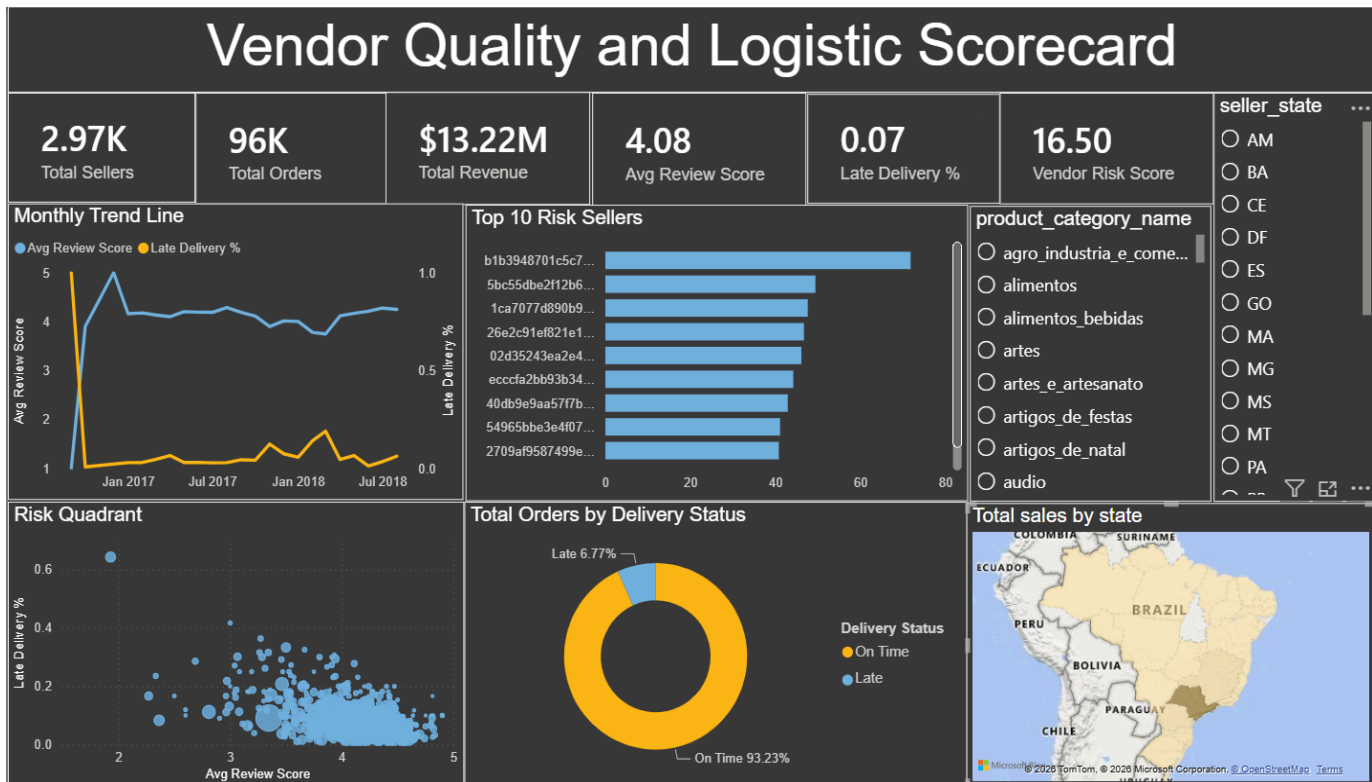
The Vendor Risk Score (0–100, higher = worse) blends rating risk (60%) and lateness risk (40%). Risk Score (Qualified) returns BLANK for sellers under 10 orders, excluding them from rankings automatically.

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Dashboard

Page 1 — Executive Overview

KPI row (2.97K sellers, 96K orders, \$13.22M revenue, 4.08 avg review, 7% late, 16.5 risk score), monthly trend, Top-10 risk sellers, risk quadrant scatter, on-time/late donut (93.23% / 6.77%), Brazil sales map, state & category slicers.



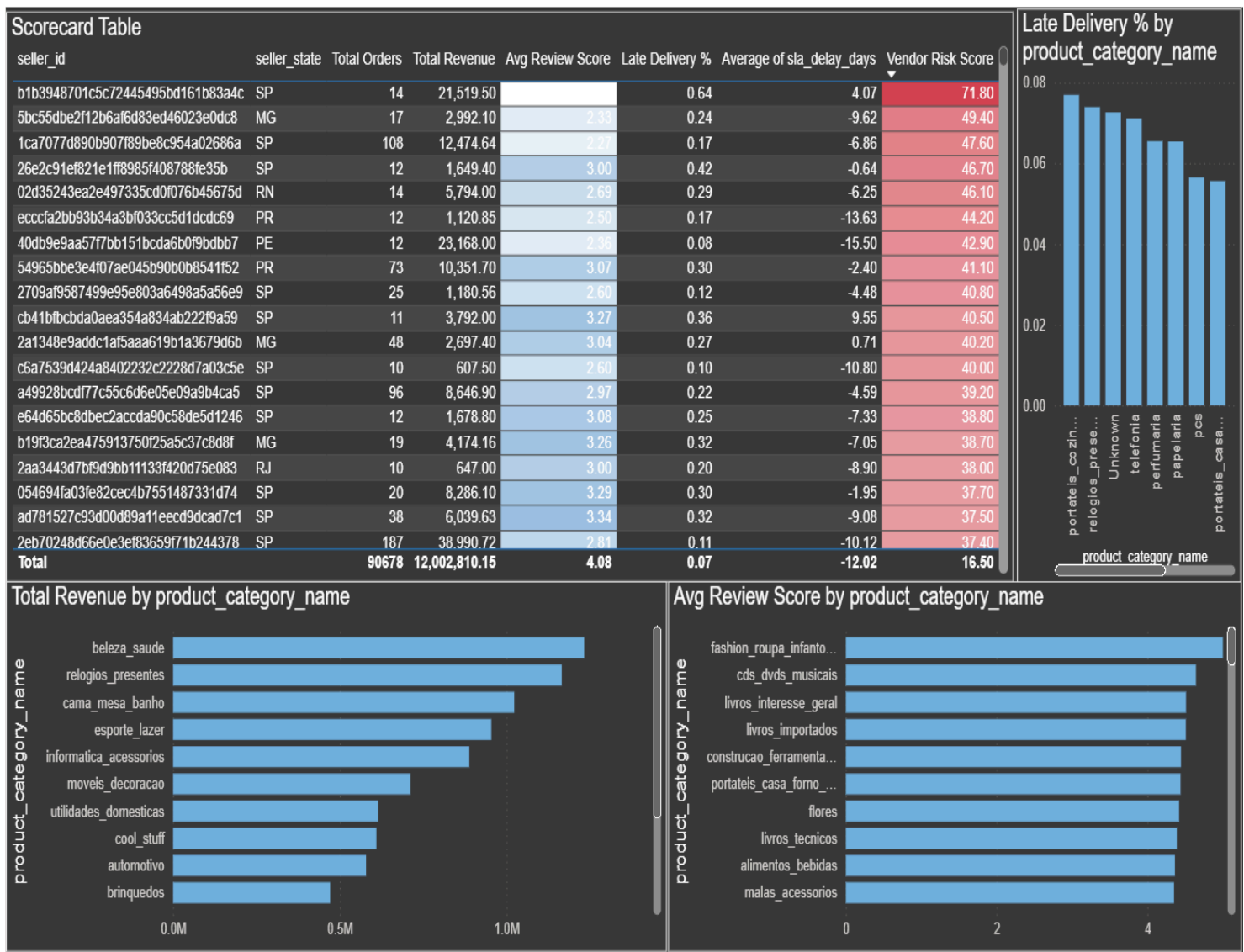
The overview answers "how healthy is the platform?" in one screen. The KPI row gives the state of the platform (2.97K sellers, 96K orders, \$13.22M revenue, 4.08 rating, 7% late, 16.5 risk). The monthly trend shows rating and lateness over time. The Top 10 Risk Sellers bar is the action list — the same sellers operations should call first. The Risk Quadrant plots every qualified seller by rating (x), late % (y), and revenue (bubble size): bad sellers cluster top-left, and a big bubble there is urgent. The delivery status donut shows 93.23% on-time vs

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6.77% late, and the sales map shows the São Paulo concentration. State and category slicers filter the entire page.

Page 2 — Seller Scorecard & Category Analysis

Ranked seller scorecard with conditional formatting on review score and risk score; Late Delivery % by category; Total Revenue by category; Avg Review Score by category.



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The scorecard table is the operational core: one row per qualified seller with orders, revenue, average review (blue scale), late %, average days vs promise, and the red-scaled Vendor Risk Score, sorted worst-first. Around it, three category charts explain where problems live: Late Delivery % by category (portable kitchen appliances, watches & gifts, and telephony worst at ~6–8%), Revenue by category (health & beauty leads at ~R\$1.2M), and Avg Review Score by category (books, CDs and kids' fashion best at 4.4+).

Business Recommendations & Expected Impact

Assumptions: Average order value is approximately **R\$138** (R\$13.22M across 96K orders). Around **6,500 orders (6.77%)** are delivered late. The estimated business impact below is directional and intended to help prioritize improvement opportunities.

1. Launch a Seller Probation Program

Place sellers with a **risk score above 40** and at least **10 completed orders** into a 60-day improvement program focused on delivery performance. Sellers who fail to improve can be demoted in search results or removed from the platform. The dashboard's **Top 10 High-Risk Sellers** provides the initial list.

Expected impact: Reducing the Top 10 sellers' late delivery rates from **30–64%** to the platform average of **7%** could prevent **800–1,200 late deliveries per year**, protecting an estimated **R\$110K–165K** in revenue while improving customer satisfaction.

Track: Risk score distribution over time.

2. Improve Delivery Time Estimates

Orders are delivered **about 12 days earlier than the estimated delivery date on average**, suggesting delivery promises are more conservative than necessary. Shorten estimated delivery times for consistently reliable sellers while keeping longer estimates only for sellers with higher delivery risk.

Expected impact: A **1% increase in conversion rate** at the current sales volume could generate roughly **R\$132K in additional annual GMV**.

Track: Conversion rate and late delivery percentage after updating delivery estimates.

3. Focus Logistics Improvements on High-Risk Categories

Since **93% of orders already arrive on time**, platform-wide logistics changes would have limited impact. Instead, prioritize categories with the highest late delivery rates, such as portable kitchen appliances, watches & gifts, and telephony, along with long-distance shipping routes.

Expected impact: Bringing the worst-performing categories closer to the 5% late-delivery rate achieved by the best categories could eliminate several hundred late deliveries each year.

Track: Late delivery percentage by product category.

4. Respond Quickly to 1-Star Reviews

Create a workflow where every **1-star review** receives a seller response within 48 hours. Monitor each seller's 1-star review rate monthly to identify recurring issues and measure improvement.

Expected impact: Retaining even a small percentage of dissatisfied customers can increase repeat purchases at

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a much lower cost than acquiring new customers.

Track: 1-star review rate and customer win-back rate.

5. Expand the Seller Base Beyond São Paulo

Sales are heavily concentrated in **São Paulo**, resulting in longer shipping distances to customers in the North and Northeast. Recruiting more regional sellers can reduce shipping times and improve delivery reliability in those areas.

Expected impact: A stronger regional seller network can shorten delivery routes, reduce late deliveries and freight costs, and improve customer satisfaction in underserved regions.

Track: Late delivery percentage and average delivery time by customer region.